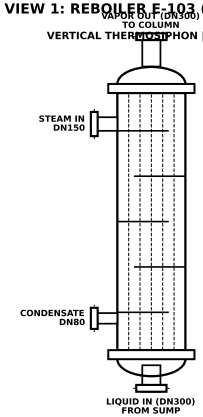
HEAT EXCHANGER FABRICATION DETAILS & 3D ISOMETRIC PLANT ASSEMBLY

THERMOSIPHON REBOILER (TEMA BEM) | CONDENSER (TEMA AEL) | ISOMETRIC SPATIAL LAYOUT

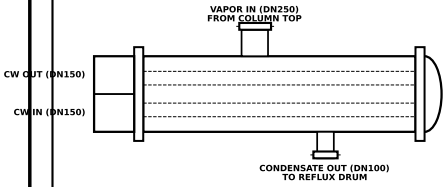
VIEW 1: REBOILER E-103 (TEMA BEM)

VERTICAL THERMOSIPHON | 3.24 MW



VIEW 2: CONDENSER E-102 (TEMA AEL)

HORIZONTAL SHELL & TUBE | 2.88 MW (CW)



VIEW 3: 3D ISOMETRIC PLANT ASSEMBLY & STRUCTURAL INTEGRATION

AXONOMETRIC PERSPECTIVE VIEW SHOWING COLUMN, LOOP & SKID

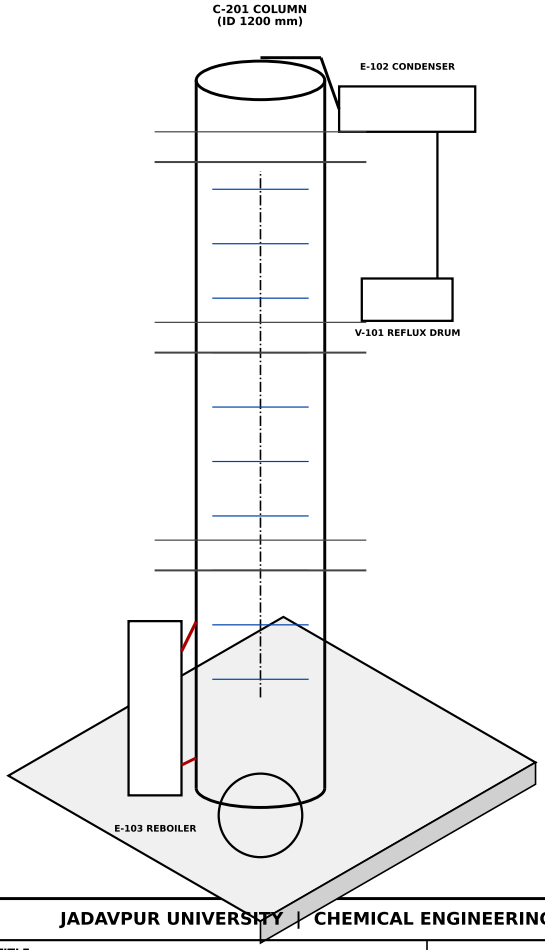


TABLE 1: HEAT EXCHANGER MECHANICAL & TEMA DESIGN DATA

SPECIFICATION	REBOILER (E-103)	CONDENSER (E-102)
TEMA Classification	BEM (Fixed Tubesheet)	AEL (Removable Bundle)
Heat Duty	3.24 MW (2.79 Gcal/h)	2.88 MW (2.48 Gcal/h)
Heat Transfer Surface Area	92.5 m2	145.0 m2
Shell Diameter (ID)	500 mm	600 mm
Tube Length	2500 mm (Vertical)	3000 mm (Horizontal)
Tube OD x Wall Thickness	19.05 mm (3/4") x 1.65 mm	19.05 mm (3/4") x 1.65 mm
Tube Pitch & Arrangement	25.4 mm Triangular (30 deg)	23.8 mm Triangular (30 deg)
Tube Count / Material	180 Tubes / 304L SS	320 Tubes / 304L SS
Shell Material	SA-516 Gr 70 / 304L Clad	SA-516 Gr 70 Carbon Steel
Design Pressure (Shell/Tube)	0.50 / 0.35 MPa(g)	0.40 / 0.60 MPa(g)
Design Temp. (Shell/Tube)	160 C / 130 C	110 C / 65 C

TABLE 2: PIPING SPECIFICATION & VELOCITY CRITERIA

SERVICE LINE	NOMINAL SIZE	SPEC / RATING	FLOW REGIME
Overhead Vapor	**DN250 (10")**	304L SS CL 150 RF	Vapor (22.5 m/s)
Reflux Return	**DN100 (4")**	304L SS CL 150 RF	Liquid (1.4 m/s)
Distillate Run-down	DN80 (3")	304L SS CL 150 RF	Liquid (1.1 m/s)
Reboiler Liquid Feed	**DN300 (12")**	304L SS CL 150 RF	Saturated Liquid
Reboiler Vapor Return	**DN300 (12")**	304L SS CL 150 RF	Two-Phase Mist
Column Bottoms Effluent	DN80 (3")	304L SS CL 150 RF	Hot Liquid (1.2 m/s)
Reboiler LP Steam	**DN150 (6")**	Carbon Steel CL 150	Dry Saturated Steam

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
REBOILER & CONDENSER DETAILS & 3D ISOMETRIC ASSEMBLY	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	AS NOTED
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	JU/ChE/ED/DWG-06	01 / 09 / 2026

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA